

Xiaopu Wang

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EDUCATION

Pennsylvania State University

Ph.D. in Statistics

2024 – Present
University Park, PA

University of Science and Technology of China (USTC)

M.S. in Statistics

2021 – 2024

Hefei, China

Nankai University

B.S. in Mathematics

2017 – 2021

Tianjin, China

RESEARCH EXPERIENCE & PUBLICATIONS

ICML 2026 • first author

2024 – 2026

Beyond Heuristic Tuning: Power-Calibrated LLM Watermarking

Wang, X., He, Z., Liu, C., Li, R.

- Derived closed-form relationships linking logit-based watermark parameters to detection power and KL distortion, replacing heuristic tuning with a constrained optimization procedure.
- Benchmarked across GPT-2, OPT, Pythia, and Gemma-2 9B on C4, LFQA, and Wikipedia.

Nature Communications 2025 • co-first author

2021 – 2025

UKB-MDRMF: A Multi-Disease Risk and Multimorbidity Framework Based on UK Biobank Data

*Jiang, Y., *Zhao, B., *Wang, X., *Tang, B., et al.

- Built the end-to-end cleaning, feature-engineering, and survival-analysis pipeline over UK Biobank (~500K participants) for multi-disease risk prediction.
- Improved atrial fibrillation AUC by **0.05** and survival c-index by **0.15** (CVD) and **0.09** (CHD) over prior models.

Statistics and Its Interface 2024 • first author

2021 – 2024

Copy Number Variation Detection Based on Constraint Least Squares

Wang, X., Wang, X., Zhang, A., Wen, C. 17(1), 27–37.

- Designed a constrained least-squares formulation with an alternating minimization algorithm; achieved the lowest estimation error across every simulation setting and was the only method to recover all change points under heavy-tailed noise.

Artificial Intelligence in Medicine 2025

2021 – 2024

CATI: A Medical Context-Enhanced Framework for Diagnosis Code Assignment

Shen, Y., Wang, J., ..., Wang, X., et al.

- Contributed to a context-enhanced model that propagates information across hierarchically related diagnosis codes on UK Biobank records.

Manuscript in preparation

2026 – Present

UniU: A Unified Cross-Model Watermark for LLM Text Detection

Wang, X., et al.

- One watermarking scheme that detects across different models and tokenizers; publicly verifiable from the text and a key alone, with no model artifacts required.

Manuscript in preparation

2025 – Present

Marked Hawkes Processes for Recurrent Events in Electronic Health Records

Wang, X., et al.

- Building marked temporal point-process (deep Hawkes) models for disease risk and patient trajectories; own the cohort-construction, preprocessing, and evaluation pipelines on MIMIC-IV.

TEACHING

Teaching Assistant

2024 – Present

Pennsylvania State University, Department of Statistics

University Park, PA

- STAT 414 (Introduction to Probability Theory) and STAT 415 (Introduction to Mathematical Statistics).
- STAT 200 (Elementary Statistics): led and lectured lab sections.

TECHNICAL SKILLS

Programming: Python, R, C++, SQL

ML & Deep Learning: PyTorch, scikit-learn; gradient boosting, deep architectures, generative models

LLMs & NLP: LLM watermarking, prompt engineering, retrieval-augmented generation

Theory: Strong mathematical statistics background

Biomedical Datasets: UK Biobank, MIMIC-IV, All of Us

Tools: Linux/shell, Git, $\text{\LaTeX}$, Mathematica